4.5 Multiplication and Division in Base Systems

Figure 4.6 The processes for multiplication and division are the same for arithmetic in any bases. (credit: modification of work “NCTR Intern Claire Boyle” by Danny Tucker/U.S. Food and Drug Administration, Public Domain)

Learning Objectives

After completing this section, you should be able to:

1. Multiply and divide in bases other than 10.
2. Identify errors in multiplying and dividing in bases other than 10.

Just as in Addition and Subtraction in Base Systems, once we decide on a system for counting, we need to establish rules for combining the numbers we’re using. This includes the rules for multiplication and division. We are familiar with those operations in base 10. How do they change if we instead use a different base? A larger base? A smaller one?

In this section, we use **multiplication and division in bases** other than 10 by referencing the processes of base 10, but applied to a new base system.

Multiplication in Bases Other Than 10

Multiplication is a way of representing repeated additions, regardless of what base is being used. However, different bases have different addition rules. In order to create the multiplication tables for a base other than 10, we need to rely on addition and the addition table for the base. So let’s look at multiplication in base 6.

Multiplication still has the same meaning as it does in base 10, in that 4×6 is 4 added to itself six times, $4 \times 6 = 4 + 4 + 4 + 4 + 4 + 4$.

So, let’s apply that to base 6. It should be clear that 0 multiplied by anything, regardless of base, will give 0, and that 1 multiplied by anything, regardless of base, will be the value of “anything.”

Step 1: So, we start with the table below:

*	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4			
3	0	3				
4	0	4				
5	0	5				

Step 2: Notice $2 \times 2 = 4$ is there. But we didn't hit a problematic number there (4 works fine in both base 10 and base 6). But what is 2×3? If we use the repeated addition concept, $2 \times 3 = 2 + 2 + 2 = 4 + 2$. According to the base 6 addition table (Table 4.4), $4 + 2 = 10$. So, we add that to our table:

*	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	10		
3	0	3	10			
4	0	4				
5	0	5				

Step 3: Next, we need to fill in 2×4. Using repeated addition, $2 \times 4 = 2 + 2 + 2 + 2 = 10 + 2 = 12$ (if we use our base 6 addition rules). So, we add that to our table:

*	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	10	12	
3	0	3	10			
4	0	4	12			
5	0	5				

Step 4: Finally, $2 \times 5 = 2 + 2 + 2 + 2 + 2 = 12 + 2 = 14$. And so we add that to our table:

*	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	10	12	14
3	0	3	10			
4	0	4	12			
5	0	5	14			

Step 5: A similar analysis will give us the remainder of the entries. Here is 4×5 demonstrated:

$4 \times 5 = 4 + 4 + 4 + 4 + 4 = 12 + 12 + 4 = 24 + 4 = 32.$

This is done by using the addition rules from Addition and Subtraction in Base Systems, namely that $4 + 4 = 12$, and then applying the addition processes we've always known, but with the base 6 table (Table 4.4). In the end, our multiplication table is as follows:

*	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	10	12	14
3	0	3	10	13	20	23
4	0	4	12	20	24	32
5	0	5	14	23	32	41

Table 4.8 Base 6 Multiplication Table

Notice anything about that bottom line? Is that similar to what happens in base 10?

To summarize the creation of a multiplication in a base other than base 10, you need the addition table of the base with which you are working. Create the table, and calculate the entries of the multiplication table by performing repeated addition in that base. The table needs to be drawn only the one time.

EXAMPLE 4.38

Creating a Multiplication Table for a Base Lower Than 10

Create the multiplication table for base 7.

Solution

Step 1: Let's apply the process demonstrated and outlined above to find the base 7 multiplication table. It should be clear that 0 multiplied by anything, regardless of base, will give 0, and that 1 multiplied by anything, regardless of base, will be the value of "anything." So, we start with the table below:

*	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6
2	0	2	4	6			
3	0	3	6				
4	0	4					
5	0	5					
6	0	6					

Step 2: Notice $2 \times 2 = 4$ is there. But we didn't hit a problematic number there (4 works fine in both base 10 and base 6). The same is true for 2×3 and 3×2, which equal 6. But what is 2×4? If we use the repeated addition concept,

$2 \times 4 = 2 + 2 + 2 + 2 = 6 + 2$. According to the base 7 addition table in the solution for Example 4.29, $6 + 2 = 11$. So, we add that to our table:

*	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6
2	0	2	4	6	**11**		
3	0	3	6				
4	0	4	**11**				
5	0	5					
6	0	6					

Step 3: Next, we need to fill in 2×5. Using repeated addition, $2 \times 5 = 2 + 2 + 2 + 2 + 2 = 11 + 2 = 13$ if we use our base 7 addition rules. So, we add that to our table:

*	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6
2	0	2	4	6	11	**13**	
3	0	3	6				
4	0	4	11				
5	0	5	**13**				
6	0	6					

Step 4: Finally, $2 \times 6 = 2 + 2 + 2 + 2 + 2 + 2 = 13 + 2 = 15$. And so we add that to our table:

*	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6
2	0	2	4	6	11	13	**15**
3	0	3	6				

4	0	4	11				
5	0	5	13				
6	0	6	**15**				

Step 5: A similar analysis will give us the remainder of the entries. Here is $4_7 \times 5_7$ demonstrated:

$$4_7 \times 5_7 = 4_7 + 4_7 + 4_7 + 4_7 + 4_7 = 11_7 + 11_7 + 4_7 = 22_7 + 4_7 = 26_7$$

This is done by using the addition rules from Addition and Subtraction in Base Systems, namely that $4_7 + 4_7 = 11_7$ and then applying the addition processes we've always known, but with the base 7 table in the solution for Example 4.29. Using those addition rules, the rest of the table is given below:

*	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6
2	0	2	4	6	11	13	15
3	0	3	6	12	15	21	24
4	0	4	11	15	22	26	33
5	0	5	13	21	26	34	42
6	0	6	15	24	33	42	51

YOUR TURN 4.38

1. Create the multiplication table for base 4.

EXAMPLE 4.39

Creating a Multiplication Table for a Base Higher Than 10

Create the multiplication table for base 12.

Solution

Let's apply the repeated addition to base 12. Here is $7_{12} \times 9_{12}$ demonstrated:

$$7_{12} \times 9_{12} = 7_{12} + 7_{12} + 7_{12} + 7_{12} + 7_{12} + 7_{12} + 7_{12} + 7_{12} + 7_{12} = 12_{12} + 12_{12} + 12_{12} + 12_{12} + 7_{12} = 48_{12} + 7_{12} = 53_{12}$$

This is done by using the addition rules from Addition and Subtraction in Base Systems, namely that $7_{12} + 7_{12} = 12_{12}$ and then applying the addition processes we've always known, but with the base 12 table in the solution for Example 4.31. Using those addition rules, the rest of the table is given below:

*	0	1	2	3	4	5	6	7	8	9	A	B
0	0	0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9	A	B
2	0	2	4	6	8	A	10	12	14	16	18	1A
3	0	3	6	9	10	13	16	19	20	23	26	29
4	0	4	8	10	14	18	20	24	28	30	34	38
5	0	5	A	13	18	21	26	2B	34	39	42	47
6	0	6	10	16	20	26	30	36	40	46	50	56
7	0	7	12	19	24	2B	36	41	48	53	5A	65
8	0	8	14	20	28	34	40	48	54	60	68	74
9	0	9	16	23	30	39	46	53	60	69	76	83
A	0	A	18	26	34	42	50	5A	68	76	84	92
B	0	B	1A	29	38	47	56	65	74	83	92	A1

Table 4.9 Base 12 Multiplication Table

YOUR TURN 4.39

1. Create the multiplication table for base 14.

The multiplication table in base 2 below is as minimal as the addition table in the solution for Table 4.6. Since the product of 1 with anything is itself, the following multiplication table is found.

*	0	1
0	0	0
1	0	1

Table 4.10 Base 2 Multiplication Table

As with the addition table, we can use the multiplication tables and the addition tables to perform multiplication of two numbers in bases other than base 10. The process is the same, with the same carry rules and placeholder rules.

EXAMPLE 4.40

Multiplying in a Base Lower Than 10

1. Calculate $45_6 \times 24_6$.
2. Calculate $101_2 \times 110_2$.

Solution

1. **Step 1:** Use the base 6 multiplication table (Table 4.8) and, when necessary, the base 6 addition table (Table 4.4).

   Set up this calculation using columns:

	4	5
x	2	4

Step 2: Multiply the 1s digits, 5 and 4, using the base 6 multiplication table (Table 4.8). There we see the result is 32_6. So, we enter the 2 and carry the 3.

	3	
	4	5
x	2	4
		2

Step 3: So, now we multiply the 4 and the 4, then add the 3 (just as you would do if multiplying two base 10 numbers!). $4_6 \times 4_6 = 24_6$ (from the base 6 table [Table 4.8]), then $24_6 + 3_6 = 31_6$. So, we enter the 31.

	3	
	4	5
x	2	4
3	**1**	2

Step 4: Now we move on to the 2 in the "tens" place in the bottom value. We multiply the 2_6 and the 5_6, and we get 14_6. So, we enter the 4 and carry the 1.

	1	
	4	5
$\times$	2	4
3	1	2
	4	0

Remember, on line two, you place a 0 in that far right spot, as a placeholder.

Step 5: Next up, we multiply the 2 and the 4, and then add 1. This gives us $12_6 + 1_6 = 13_6$. We enter those on that second line.

		1	
		4	5
	x	2	4
	3	1	2
1	**3**	4	0

Step 6: Now we add down the columns.

		1	
		4	5
	x	2	4
1	3	1	2
1	3	4	0
2	0	5	2

Step 7: The 3 and the 3 add to 10 in base 6, so we enter the 0 and carry the 1. We now have the result:

$45_6 \times 24_6 = 2052_6$.

2. **Step 1:** Use the base 2 multiplication table (Table 4.10) and, when necessary, the base 2 addition table in the solution for Example 4.29. Set up this calculation using columns:

	1	0	1
x	1	1	0

Step 2: Using the pattern established above, and the processes from multiplication from base 10, we find the following:

			1	0	1
x			1	1	0
			0	0	0
		1	0	1	
	1	0	1		

Step 3: Adding down the columns results in the following:

			1	0	1
x			1	1	0
			0	0	0
		1	0	1	
	1	0	1		
	1	1	1	1	0

So, $101_2 \times 110_2 = 11110_2$.

YOUR TURN 4.40

1. Calculate $43_6 \times 52_6$.
2. Calculate $11101_2 \times 11_2$.

Summarizing the process of multiplying two numbers in different bases, the multiplication table is referenced. Using that table, the multiplication is carried out in the same manner as it is in base 10. The addition rules for the base will also be referenced when carrying a 1 or when adding the results for each digit's multiplication line.

EXAMPLE 4.41

Multiplying in a Base Higher Than 10

Calculate $3\text{A}_{12} \times 74_{12}$.

Solution

Step 1: Use the base 12 multiplication table in the solution for Example 4.39 and, when necessary, the base 12 addition table in the solution for Table 4.7. Set up this calculation using columns:

	3	A
×	7	4

Step 2: First, the 4 is multiplied by 3A, resulting in the first line.

	3	A
×	7	4
1	**3**	**4**

Step 3: Now we move on to the 7 in the "tens" place in the bottom value.

		5	
		3	A
	x	7	4
	1	3	4
2	**2**	**A**	**0**

Step 4: Now we add down the columns.

		3	A
	x	7	4
	1	3	4
2	2	A	0
2	**4**	**1**	**4**

Step 5: The 3 and the A add to 11 in base 12, so we enter the 1 and carry the 1.

We now have the result: $3\text{A}_{12} \times 74_{12} = 2414_{12}$.

YOUR TURN 4.41

1. Calculate $\text{B}3_{12} \times 47_{12}$.

Division in Bases Other Than 10

Just as with the other operations, **division in a base** other than 10, the process of division in a base other than 10 is the same as the process when working in base 10. For instance, $72 \div 9 = 8$ because, we know that $9 \times 8 = 72$. So, for many division problems, we are simply looking to the multiplication table to identify the appropriate multiplication rule.

EXAMPLE 4.42

Dividing with a Base Other Than 10

1. Calculate $14_6 \div 5_6$.
2. Calculate $5\text{A}_{12} \div 7_{12}$

Solution

1. Looking at the multiplication table for base 6 (Table 4.8), we see that $5_6 \times 2_6 = 14_6$. Using that, we know that $14_6 \div 5_6 = 2_6$.
2. Looking at the multiplication table for base 12 in the solution for Example 4.39, we see that $7_{12} \times \text{A}_{12} = 5\text{A}_{12}$. Using that, we know that $5\text{A}_{12} \div 7_{12} = \text{A}_{12}$.

YOUR TURN 4.42

1. Calculate $10_6 \div 3_6$

2. Calculate $50_{12} \div A_{12}$.

Errors in Multiplying and Dividing in Bases Other Than Base 10

The types of errors encountered when multiplying and dividing in bases other than base 10 are the same as when adding and subtracting. They often involve applying base 10 rules or symbols to an arithmetic problem in a base other than base 10. The first type of error is using a symbol that is not in the symbol set for the base.

EXAMPLE 4.43

Identifying an Illegal Symbol in a Base Other Than Base 10

Explain the error in the following calculation, and determine the correct answer:

$$4_6 \times 2_6 = 8_6$$

Solution

Since the problem is in base 6, the symbol set available is 0, 1, 2, 3, 4, and 5. The 8 in the answer is clearly not a legal symbol for base 6. Looking back to the base 6 multiplication table (Table 4.8), we see that $4_6 \times 2_6 = 12_6$.

YOUR TURN 4.43

1. Explain the error in the following calculation and determine the correct answer: $13_4 \times 21_4 = 54_4$

The second type of error is using a base 10 rule when the numbers are not in base 10. For instance, in base 17, $6_{17} \times 9_{17} = 54_{17}$ would be incorrect, even though in base 10, $6 \times 9 = 54$. That rule doesn't apply in base 17.

EXAMPLE 4.44

Identifying an Error in Arithmetic in a Base Other Than Base 10

Explain the error in the following calculation. Determine the correct answer:

$$18_{12} \times 7_{12} = 126_{12}$$

Solution

If this problem was a base 10 problem, this would be the correct answer. However, in base 12, $8_{12} \times 7_{12}$ is not 56, but is instead 48. To correct this error, carefully use the multiplication table for base 12 (Table 4.9). If properly used, the correct answer would be $18_{12} \times 7_{12} = B8_{12}$.

YOUR TURN 4.44

1. Explain the error in the following calculation. Determine the correct answer: $49_{14} \times 9_{14} = 441_{14}$

Check Your Understanding

28. To create the multiplication table for a given base, what should be used?
29. What are the differences between multiplying in base 10 and multiplying in a different base?
30. When dividing in a base other than base 10, what table is referenced?
31. Compute $24_6 \times 53_6$.
32. Compute $32_{14} \div 4_{14}$.
33. How do you know an error has occurred in a base 5 multiplication question if the answer obtained was 28_5?
34. What are two common ways to determine an error is committed when computing in s base other than base 10?

SECTION 4.5 EXERCISES

For the following exercises, create the multiplication table for the given base.

1. base 5
2. base 3
3. base 16 (Hint: Use the digits 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E, F.)
4. base 2

For the following exercises, perform the indicated base 6 operation.

5. $4_6 \times 3_6$
6. $14_6 \times 5_6$
7. $31_6 \times 3_6$
8. $43_6 \times 34_6$
9. $532_6 \times 23_6$
10. $254_6 \times 143_6$
11. $20_6 \div 3_6$
12. $23_5 \div 5_6$

For the following exercises, perform the indicated base 12 operation.

13. $5_{12} \times 6_{12}$
14. $3_{12} \times \mathrm{A}_{12}$
15. $34_{12} \times 7_{12}$
16. $76_{12} \times \mathrm{B}_{12}$
17. $59_{12} \times 1\mathrm{A}_{12}$
18. $\mathrm{A}1_{12} \times 36_{12}$
19. $53_{12} \div 9_{12}$
20. $2\mathrm{B}_{12} \div 7_{12}$

21. Explain two ways to detect an error in arithmetic in bases other than base 10.

22. Explain the error in the following calculation: $15_{12} \times 7_{12} = 105_{12}$

23. Explain the error in the following calculation: $45_8 \times 6_8 = 94_8$.

24. In base 10 multiplication, there are 100 multiplication rules plus a rule for carrying a number. How many multiplication rules are there for base 6?

25. In base 10 multiplication, there are 100 multiplication rules plus a rule for carrying a number. How many multiplication rules are there for base 14?

26. In base 10 multiplication, there are 100 multiplication rules plus a rule for carrying a number. How many multiplication rules are there for base 2?

27. Consider the answers from Exercise 24 and Exercise 26. Which base do you think would be more efficient: base 10, base 6, or base 2?

For the following exercises, use the multiplication table that you created from Exercise 4 to perform the indicated base 2 operations.

28. $11_2 \times 11_2$
29. $101_2 \times 10_2$
30. $11011_2 \times 1011_2$
31. $1011_2 \times 1010101_2$

32. Convert 1011_2 and 1010101_2 to base 10. Then multiply those base 10 numbers. Next, convert the answer you got for Exercise 31 to base 10. Do these numbers match?

For the following exercises, use the multiplication table that you created from Exercise 3 to perform the indicated base 16 operations.

33. $19_{16} \times 5_{16}$
34. $3\mathrm{B}_{16} \times \mathrm{A}_{16}$
35. $25_{16} \times 16_{16}$
36. $\mathrm{C}_{16} \div 4_{16}$

For the following exercises, explain how you know an error was committed without performing the operation in the given base.

37. $43_5 \times 32_5 = 126_5$

38. $5_{14} \times 9_{14} = 45_{14}$